

Why can equal results lead to different scientific futures

AI now participates in mathematical research and sustained scientific work. How do humans and agents, through historically situated joint practice, form capacities that change their subsequent conditions of problem solving? Mathematicians' concerns about understanding, OpenAI's Navier–Stokes process account and persistent-agent workflows make this question concrete.^{1–3} We examine the formation, adoption and reconstruction of relations connecting problems, methods, evidence and responsibilities. A finite construction establishes that equality of a specified current readout need not determine the correction required next: correction from that readout alone has a positive lower bound, removed by a compact state-aware rule. Archived agent records show feedback-driven programme revision and reuse under changed targets. In one revision, a non-zero component cancels an inherited-state term and improves the whole, relinquishing an already exact subresult. Separately, a selected known-proof completion turns shared declarations into a premise used downstream: with explicit researcher-written guidance, the third candidate is accepted in the saved compilation record, whereas the comparison remains incomplete. Convergence, unchanged versions and execution-dependent contrasts remain part of these records, which do not establish a general history effect. Together, the mathematical separation, sufficient repair and traceable instances distinguish obtaining results from forming and sustaining the relations that make them usable for further discovery.

Introduction

As artificial intelligence becomes more capable and enters scientific work more directly, getting a result and sustaining the capacity to develop it become distinct practical concerns. Mathematicians' joint statement stresses understanding, method transmission, training and new questions as purposes of problem solving.¹ OpenAI's published Navier–Stokes (NS) account describes intermediate Euler results being incorporated into subsequent NS work and group insights being reorganised for further investigation.² In persistent Codex workflows, working files, notes, context transitions and tool feedback

27 make retaining and revising earlier work concrete operations.³ These settings expose a common question:
28 when people and AI produce useful results, what conditions do they also form for inquiry to continue?

29 Our central question is **how humans and agents, through historically situated joint practice, form**
30 **capacities that change their own conditions of problem solving.** Our candidate answer concerns the
31 relations connecting problems, methods, evidence and responsibilities: research can form the conditions
32 for subsequent research, and these conditions can themselves require maintenance or reconstruction. The
33 same declared result need not determine how it can be checked, corrected or developed next. We establish
34 one precise limitation on a result-centred description: equality of a specified current readout can leave an
35 unavoidable discrepancy between the corrections required by different successor states. Preserving a
36 compact, relevant state removes that discrepancy. “Equal” names a specified product, observation or
37 current function, not a complete causal state; “scientific futures” names subsequent duties within stated
38 tools, permissions and resources. Compact preservation and legitimate reconstruction are positive
39 sufficient cases.

40 This distinction has established foundations. Proofs convey methodological knowledge beyond
41 statements, and productivity need not coincide with understanding.^{4,5} Retrieval-assisted proving studies
42 the availability of premises; truth-maintenance systems maintain reasons and revise dependent beliefs.^{6,7}
43 Predictive representations characterise state through future action-dependent consequences, and
44 equivalent expressive resources can have different optimisation dynamics.^{8,9} Organisational routines link
45 situated performances with patterns they preserve and change.¹⁰ These accounts already address use,
46 change and relationships. Our explanatory contribution is to organise the production, adoption and
47 reconstruction of research conditions as a connected object of comparison: a usable method interface, an
48 evidential obligation or an allocation of checking can be an outcome of earlier work and a condition of
49 later work. We operationalise this object by separating what is available from what is actually used, what
50 remains sufficient from what must be reconstructed, and a specified output from the state required for
51 continuation. The finite witness and the two kinds of process record make different parts of these

52 distinctions inspectable. Accordingly, an output comparison and a comparison of what can be done next
 53 need not hold the same things fixed.

54 Recent studies make this distinction timely. Research on model warmth tests a relation between desirable
 55 conversational properties rather than assuming their compatibility.¹¹ Model-collapse research examines
 56 how generated outputs alter later training conditions.¹² Paper2Agent converts papers and computational
 57 resources into interactive, executable research objects.¹³ Our concern is complementary: once a result or
 58 tool is available, how is its applicability established, used and reconstructed when its consequences
 59 expose a new requirement? We do not claim a newly discovered law of memory or proof application. We
 60 make this relation inspectable through a finite non-sufficiency result, a constructive repair, actual
 61 mathematical revision and successor-use records, and a known-proof application whose generated
 62 premise is consumed downstream. The mathematical result specifies a distinction between comparisons;
 63 the coefficient records make revision and changed-target reuse inspectable; and the Lean record shows a
 64 generated premise actually being consumed.

65 **Practice makes future conditions**

66 Consider a bounded research system containing people, agents, records, tools and the relevant
 67 environment. Let x_t denote its causally relevant state and $P_t = M(x_t)$ a specified present product or
 68 observation. Its recorded history H_t is evidence about how it reached that state, not a substitute for the
 69 state itself. A research arrangement links a problem to an admissible method, a method to evidence, and
 70 evidence to a participant or procedure authorised to use, challenge or revise it. These links have concrete
 71 carriers: parameter bindings, proof interfaces, test provenance, access rules, methodological explanations
 72 and assignments of responsibility. They may be explicit, distributed across participants or only partly
 73 recorded.

74 Four comparisons must therefore be separated. Product equivalence fixes a declared output or
 75 observation. Record equivalence fixes selected stored material. Complete causal-state equivalence fixes
 76 the relevant system and environment, rather than just their documentation. Finally, operational sufficiency

77 is relative to a future regime F : specified duties, tools, permissions, resources and a declared family of
 78 continuation policies. Two states are equivalent under F when their relevant future outcomes and costs
 79 have the same law under those matched policies. A proof, an error value or a list of lemmas can be equal
 80 while the other comparisons are not. If complete causal states and transition rules are equal, an unencoded
 81 historical force cannot explain different futures.

82 A research episode begins with an arrangement that makes some actions possible: a problem formulation,
 83 methods that participants know how to use, evidence with an assigned scope, and responsibilities for
 84 checking it. Joint work produces more than a saved answer when it changes this arrangement. A tested
 85 estimate may justify choosing a new route; a proof may expose a reusable parameter substitution; a failed
 86 derivation may identify a missing hypothesis. The relevant carrier can be a formula, an executable file, an
 87 explanation, a test receipt or a person's acquired method. Formation is identified by an actual construction
 88 or revision history, not inferred merely because a useful object exists.

89 Adoption is the next distinct operation. A result becomes a working condition when it constrains a real
 90 decision or is instantiated as the premise of another inference. The resulting action produces
 91 consequences: a test confirms a limited range, a compiler rejects a binding, or a new application reveals
 92 that the evidence did not cover the intended case. Such feedback can leave the method unchanged, narrow
 93 its scope, alter an interface, or change who must check the claim. The changed arrangement becomes a
 94 reproduced research condition only when it participates in another action. Figure 1 distinguishes the
 95 established separation from its recorded operational instances. This sequence explains why accumulating
 96 correct outputs and maintaining future usability can coincide, but need not do so.

97 The human organiser belongs inside this sequence. Marx's third thesis on Feuerbach challenges accounts
 98 that place the educator outside the process of change.¹⁴ Mao's account of practice centres renewed action
 99 and revision under changed conditions.¹⁵ Here these ideas determine the boundary of the research system:
 100 selection, diagnosis, methodological explanation, checking labour and their possible errors must be
 101 recorded alongside agent behaviour. They do not prove a computational mechanism. In the retained study,

human interventions altered the task and supplied an applicability strategy. Treating that contribution as free, infallible background would misattribute jointly organised performance to the model alone.

Three operations have different evidential requirements. **U1**, using an existing method, includes applying a known theorem with the right substitutions. **U2**, forming a method through practice, requires a newly established procedure or relation with an identifiable production history; constructing a particular proof object is a narrower instance than discovering a general method. **U3**, reconstructing a method under changed conditions, requires a documented change in what the method must handle, a corresponding reconstruction and evidence of subsequent use. These operations need not occur in order or in every episode. A stable classical method may remain sufficient. A never-formed condition is not a lost one; an available but unused method is not forgotten; and restoration of a previously available arrangement differs from first formation or new reconstruction.

The account yields three conditional propositions. First, if a later duty depends on an applicability relation not determined by a specified current product, product equality alone cannot warrant continuation equality; the mathematical example below provides one exact case. Second, if a preserved or reconstructed interface supplies all required observations, updates and operations within budget, a more elaborate historical carrier is unnecessary for that regime. Third, a claim that joint practice produced a capacity requires a trace of formation, actual use and relevant consequences, not just a correlation between archive size and a final score. These propositions change what a comparison holds fixed: current output, available information, adopted working relation and complete state are different controls. An intervention that changes a purportedly relevant relation but leaves the predicted consequence unchanged, without repair through another route, would narrow its proposed role. Stable classical methods and adequate single-system implementations are substantive sufficient cases, not failed controls.

Equal readouts, different corrections

The mathematical witness has a precise explanatory role: under its stated observation restriction, a present result does not determine the correction required next, whereas a compact state-aware operation does. A

127 specified present readout is insufficient for exact correction across two admissible successor states. The
 128 source NS work uses a fixed primary wave and a covariance operator; positive amplitudes and an integral
 129 matrix support later signed corrections while retaining a quadratic remainder.¹⁶ We examine its explicitly
 130 derived coefficient reduction. Let z be the actual amplitude relative to a fixed base, p the target squared
 131 amplitude, $r = p - z^2$ the residual readout and g an applied increment. Writing $u = z - 1$, the complete
 132 consequence is

$$133 \quad E = (z + g)^2 - p = (2g + g^2 - r) + 2ug. \quad (1)$$

134 The first term is the pure correction residual at the fixed base; the second contains the effect of the state
 135 already produced. Reducing the first term need not reduce their sum. Thus a continuation cannot be
 136 described solely as solving the same fresh problem repeatedly: the previous operation changes what the
 137 next one must receive.

138 Fix $0 < r \leq 1/4$ and $0 < v \leq 1/8$, and consider $z_{\pm} = 1 \pm v$ with targets $p_{\pm} = z_{\pm}^2 + r$. These two states have the
 139 same declared readout r . Any deterministic action depending only on that readout must use the same g ,
 140 giving

$$141 \quad E_{\pm} = g^2 + 2(1 \pm v)g - r, E_+ - E_- = 4vg. \quad (2)$$

142 Both residuals cannot vanish: that would require $g = 0$ and hence $r = 0$. More strongly,

$$143 \quad \inf_{g \in \mathbb{R}} \max(|E_+|, |E_-|) = 2v(\sqrt{1+r} - 1) > 0. \quad (3)$$

144 To see the bound, write the maximum as $|g^2 + 2g - r| + 2v|g|$. On $g \geq 0$ its minimum occurs at
 145 $g = \sqrt{1+r} - 1$; on $[-2, 0]$ it is at least r , and reflection of the remaining negative branch adds a positive
 146 offset. Supplementary Note 2 gives the piecewise proof. The restriction is explicit: the same deterministic
 147 action sees only r , whereas $z_{\pm}$ and $p_{\pm}$ differ. This is not equality of complete current function, nor an
 148 assumption that the model in the later case ignored its state.

149 Adding the relevant state gives a constructive alternative. For $r \in [-1/2, 1/2]$ and $u \in [-1/4, 1/4]$, let
 150 $z = 1 + u$ and $w = \sqrt{z^2 + r}$. The radicand is at least $1/16$. Define smooth core functions

$$151 \quad C_*(r, u) = -\frac{1}{2z}, B_*(r, u) = -\frac{1}{2z(w+z)^2}. \quad (4)$$

152 With a smooth cutoff Φ equal to one on $|r| \leq 1/4$, $|u| \leq 1/8$ and zero outside the stated outer rectangle, the
 153 operation

$$154 \quad g = \frac{r}{2} + \Phi \left[r^2 B_* + u r C_* \right] \quad (5)$$

155 satisfies $g = w - z$ and $E = 0$ in the inner region. Outside the outer rectangle it returns to the signed linear
 156 correction; the transition remains smooth. The relevant state and a compact operation thus suffice for
 157 exact local correction. Positivity and all denominators are checked on an open neighbourhood of the
 158 stated outer rectangle, with no evaluation of a square root across a negative radicand.

159 This witness gives the title a precise meaning (Fig. 2): equality of a declared present quantity can coexist
 160 with a positive lower bound for observation-restricted correction, and the separation can disappear when
 161 the required state is preserved or reconstructed. These are standard algebraic tools applied to a specified
 162 continuation problem. Extending the construction to weighted wave classes requires the source derivative,
 163 support and multiplier assumptions. Neither the local witness nor a finite numerical check independently
 164 proves the global NS theorem, universal norm improvement or the properties of arbitrary generated code.
 165 Supplementary Note 2 states this mathematical scope once alongside the preserved counterexamples.

166 **When compact inheritance is enough**

167 A compact record can preserve continuation without preserving the entire history. Suppose a compact
 168 record $c = C(x)$ supports the observations, permitted actions, successor updates and costs required by a
 169 declared family of future duties. If these laws factor through c and its successor can be updated by the
 170 same rule, a matched policy reproduces the relevant finite continuation law. This is the standard
 171 sufficient-state composition principle, proved by induction in Supplementary Note 3 and consistent with

172 predictive representations.⁸ The result includes the transitions: reproducing today's answer alone is
 173 insufficient. Legitimate rederivation also supplies a sufficient route when its construction and retrieval
 174 costs fit the specified budget.

175 The scientific problem is consequently not to maximise retained history. It is to identify what must remain
 176 actionable as the duties change. A compact interface, a conventional method or a single integrated solver
 177 can be fully sufficient; copying an entire transcript may add no value. An archive can preserve
 178 information without making it usable at the relevant time. Conversely, a summary can preserve the right
 179 conditions without preserving their entire production history. A shift to new duties can invalidate a
 180 previous sufficiency judgement, making the representation or method itself a legitimate object of further
 181 practice. This is the transition from inheritance to reconstruction, rather than an assumption of inevitable
 182 degradation.

183 **Recorded revision and continued use**

184 The coefficient records connect the distinction in equation (1) to actual formation, feedback-driven
 185 revision and subsequent use: an exact pure component can leave a whole residual through its interaction
 186 with the inherited state. Archived coefficient studies record actual revision and reuse under changed
 187 targets. Four formation episodes in an assistance comparison and eight in a version-adoption comparison
 188 produced 52 recorded stages and 104 completed model returns. Real programmes, feedback and the
 189 resulting amplitude fields were carried forward to changed targets. The common operation was
 190 $g = r/2 + \Phi(r, u)[r^2 B(r, u) + urC(r, u)]$; the agent supplied B, C , while the investigators supplied the
 191 template, cutoff, sources and duties. These observations directly instantiate equation (1), but not a
 192 complete NS construction. Two post-selected traces make the mathematical content of this continuation
 193 explicit (Fig. 3; Supplementary Note 9).

194 In one assistance episode, the initially formed rule retained $B_0 = -1/[2(1 + \sqrt{1+r})^2]$ and $C_0 = 0$. Writing
 195 $s = \sqrt{1+r}$, it gives $g = s - 1$: its pure residual is exactly zero, yet its full successor residual is $2u(s - 1)$.
 196 The agent's first revision introduced $C_1 = -1/[s(1+s)]$, changing the complete residual to $-u^2 r/(1+r)$.

197 After numerical feedback, it produced a smooth secant expression for C , while retaining B_0 . On the
 198 declared core this gives $g = \sqrt{(1+u)^2 + r} - (1+u)$ and an exactly zero complete residual. The recorded next
 199 continuation received and retained that actual programme and used its predecessor's state. Crucially, the
 200 revised pure residual is generally non-zero: it cancels the inherited-state cross term. Correct continuation
 201 here required relinquishing a locally exact subresult, not improving every component separately.

202 A second episode recorded $B = -1/8$ and successive programmes with $C = -1/2, -1/2 + u/2$, and
 203 $-1/2 + u/2 - u^2/2$. These were generated, applied and passed onward, rather than inserted retrospectively
 204 by the analyst. For $u = a\tau$ and $r = b\tau^3$, their whole residuals generally begin at orders τ^5, τ^6 and τ^7 . The
 205 later duty was evaluated on the actual inherited field, not a reset base. This is a finite instance of revision
 206 under a changed target and renewed use. It is not a new general series method: the coefficients follow a
 207 familiar expansion, and the original explanation of one remaining term contained a sign error that is
 208 distinguished from the effective programme in Supplementary Note 9.

209 These traces are process evidence, not substitutes for the original comparisons. Seven of eight version
 210 pairs had identical initial and revised code; five final responses across the two studies were empty. In the
 211 fixed outer region, any valid programme applied $r/2$, whereas an empty final response applied no update.
 212 This route dominated large positive or negative aggregate contrasts. All original denominators and
 213 outcomes are retained, including the separately authorised supplementary observation replacing an
 214 unavailable analysis slot without resolving its historical unknown request. No stable average assistance or
 215 adoption benefit was established. Earlier finite-scalar comparisons also converged after adaptation. Thus
 216 revision and renewed use were observed, whereas their general causal advantage remains unresolved.
 217 This distinction replaces neither null results with selected successes nor executed work with a claim that
 218 no continuation occurred.

219 **An available result becomes a premise**

220 A separate proof-completion record shows an existing mathematical result being made usable by a
 221 supplied consumer, at a different location in the covariance construction. The recorded task completes

222 `hclose` inside the supplied Lean theorem `compact_chart_pair_bounds`.^{17,18} The outer proof and two relevant
 223 lemma declarations were identical in the two conditions. An additional note in one condition explained
 224 how to apply those results to the current obligation. This deliberately selected, known-proof task supplies
 225 a local record of turning existing mathematics into an accepted premise; it is not a controlled instantiation
 226 of the two-state lower bound in equation (3).

227 The common mathematical assumption bounds a pulse-ratio error on an interval of length L_n . The
 228 available integral estimate expects the interval, denominator and centre in terms of R_n^2 . A second common
 229 lemma states $R_n^2 = L_n$. The obligation is therefore not to discover a missing numerical inequality, but to
 230 transport the entire dependent premise into the representation required by the integral estimate. Rewriting
 231 interval membership alone is insufficient: the conclusion obtained from that membership must also be
 232 rewritten. Figure 4 locates the exact application and its fixed downstream use.

233 After this transport, the supplied integral lemma yields

$$234 \quad |\overline{H}(P)_{ij} - H_0(\xi)_{ij}| \leq \frac{e+dc}{R_n} \leq \frac{e+d|c|}{R_n} = \frac{Q}{R_n} \leq \rho. \quad (6)$$

235 Here $d \geq 0$, $R_n > 0$, $Q = e + d|c| \geq 0$, and the last comparison is already provided. The symbol e denotes the
 236 source error coefficient, not the residual E in equation (1). No division by Q is required, so the case $Q = 0$
 237 is retained. The resulting `hclose` object is passed to the pre-written `hmargin` application, which supplies
 238 margins subsequently combined with column-scale bounds to construct `ZeroOrderBounds`. This is actual
 239 consumption of a local proof object by downstream code, not an inferred relevance relation. The pulse
 240 construction, integral estimate, compactness argument and downstream plan themselves were supplied.

241 Abstractly, a result $L : \forall \theta, A(\theta) \rightarrow C(\theta)$ becomes usable for a consumer requiring I only after an
 242 admissible parameter choice, evidence for $A(\theta)$ and any necessary transport from $C(\theta)$ to I have been
 243 supplied. This is ordinary proof composition, not a new logical mechanism. It specifies the concrete
 244 relation observed here: common declarations did not by themselves determine whether a usable instance

was constructed within the available work sequence. A correct compact record of that instantiation could be sufficient. The empirical case concerns application, whereas equations (1)–(5) concern state-dependent correction. Their correspondence is explanatory, not a claim that the two experiments or causal mechanisms are identical.

What the recorded comparison establishes

The saved record contains six returned model requests and five compiled candidates (Fig. 4). The applicability-guided condition produced two rejected candidates, then a third accepted candidate. The declarations-only condition first reached the 32,768-token output limit without a visible proof body; its next two candidates were rejected. The accepted body was mechanically inserted into the common enclosing theorem, whose saved Lean output reports the exact local type and dependencies `propext`, `Classical.choice` and `Quot.sound`. Acceptance here denotes that retained compilation record; the local target’s meaning and the status of subsequent verification are stated separately in Methods.¹⁹

Both final candidates invoked the relevant integral lemma and scale identity. The final declarations-only body converted the interval-membership hypothesis, but returned an inequality still expressed using L_n where the expected type used R_n^2 . The accepted candidate completed the representation change and the scalar comparison. The difference is therefore narrower and more informative than “one condition found the theorem and the other did not”. It identifies where an available result failed to become a consumer-ready premise. The guided condition’s earlier failures are retained alongside this success; the first truncated comparison request is not labelled a mathematically refuted proof.

The investigators selected this gap after several broader workflows had been modified or stopped. The guided note was researcher-written procedural information, not a blind summary of history; it added 332 initial input tokens. Realised computation also differed: guided requests used 25,757 input and 20,781 output tokens, compared with 12,124 and 41,009, respectively, in the comparison. Compiler feedback and subsequent inputs were consequently different. The experiment documents this local application sequence and its recorded contrast. It does not establish a general success rate, efficiency advantage or causal effect

270 isolated from extra guidance and context use. Position and presentation are plausible alternatives in light
271 of previous context-use research.²⁰ No history-retrieval or compression manipulation occurred in the final
272 protocol.

273 The proof-completion episode records use of existing methods, formation of a particular proof object and
274 feedback-based revision of that object. Its downstream consumer was supplied, so it does not by itself test
275 reconstruction for a new duty. The coefficient episodes do contain changed-target revision and renewed
276 use, under an investigator-provided interface. These complementary records support the existence of
277 particular application and reconstruction processes, not autonomous formation of every condition in the
278 theoretical cycle. Their distinct inputs, tasks and sampling prevent their treatment as independent
279 confirmations of one general effect.

280 These distinctions also prevent several different histories from receiving the same explanation. An empty
281 terminal return records no final programme, not proof that a method was never formed; a draft may
282 already exist. Identical code can still be adopted and used on an inherited state. A failed conversion of an
283 existing lemma concerns applicability at the consumer interface, not absence of the lemma. Changed-
284 target revision followed by renewed use is documented in the coefficient traces; first non-formation, an
285 available method never adopted, and restoration of a lost arrangement remain distinct theoretical
286 possibilities rather than additional observed categories in these records. Formation of the accepted proof
287 object does not measure a change in human understanding.

288 **Conditions for further discovery**

289 The account makes the formation, adoption and reconstruction of usable research conditions an explicit
290 object of explanation and evaluation. Its established mathematical separation is specific and constructive:
291 a declared current readout need not determine the correction required next, and relevant state can provide
292 a compact sufficient repair. Actual records show both a formed programme being revised and reused
293 under changed targets and an available theorem being made into a premise consumed by downstream
294 code. In the former, improving the complete result could require abandoning exactness of a component.

295 Together these contributions connect an exact distinction between comparisons to inspectable operations
296 in joint practice. They make result production and the maintenance of usable research conditions
297 distinguishable objects of inquiry, without requiring long histories to be intrinsically valuable or a new
298 force beyond ordinary state and transition descriptions.

299 For mathematical understanding and training, the finite separation changes the comparison that matters.
300 Equal assessed answers do not establish that learners can check assumptions, recognise another use or
301 formulate the next problem. Evaluation can therefore pair a current result with a later duty requiring a
302 stated relation—such as adapting a method after a hypothesis changes—and record both. The human
303 prediction is that differences appear where the necessary relation is not fixed by the assessed product and
304 cannot be reconstructed within budget. If a compact explanation or ordinary rederivation restores transfer,
305 it is a successful educational design, not a failure to demonstrate historical necessity. If matched product
306 performance also predicts all prespecified continuation duties across such interventions, the proposed
307 additional distinction is unnecessary in that setting. Testing these claims requires actual human activity;
308 Lean acceptance is evidence about a proof object, not a measure of understanding.^{1,4,5}

309 For research organisation, an intermediate result has value both as a finding and as a possible change to
310 the next research arrangement. It can redirect a route, shift the evidential responsibility between
311 participants, or justify reallocating resources. Recording only the final output obscures the labour that
312 makes these changes workable. A contribution record should therefore distinguish production,
313 applicability work, actual adoption and continuing maintenance; it should also record unsuccessful
314 coordination and corrections to human judgement. The NS process account motivates this distinction
315 without separating the effects of transfer, model changes, resources and selection.² Organisational
316 research can test this relation where duties and access can be documented. A handoff intervention can
317 track changes in checking or research routes, then test whether another carrier restores the same outcome
318 and cost. Equivalent compact handoffs demonstrate preservation of the relevant conditions; no change
319 without such restoration narrows the proposed relation's role.¹⁰

320 For persistent agents, a completed task can be evaluated by what remains possible afterwards: correction
321 under new evidence, adjustment of scope and reconstruction of a method for a changed objective. Pinned
322 Codex sources make notes, context transitions and historical access distinct operations; their availability
323 does not establish that the right conditions are preserved.³ Executable-paper systems provide another
324 useful carrier.¹³ The design question is what state or application interface must survive, who can revise it,
325 and whether the revision is used. A compact certified interface, reliable classical routine or integrated
326 solver may satisfy these requirements without a long transcript. Conversely, preserving every message
327 may still fail to make the relevant relationship actionable. These predictions require specified future
328 duties, budgets and observations of restoration; they do not license inferring alienation or loss of agency
329 from one failed proof attempt.

330 Written proofs, executable checks, preserved programmes and complete comparison records provide a
331 basis for independent reproduction and extension. The present work is responsible for its finite claims,
332 their assumptions and these traceable instances. Mathematical training can investigate how current
333 achievement relates to later checking and method transfer; organisational research can track how adoption
334 changes responsibilities and research routes; persistent-agent research can test what must remain
335 actionable for correction and adaptation. Each extension has to specify its duties and costs and retain
336 cases in which compact records, classical methods or rederivation suffice. The common question is not
337 simply how many results a system produces, but what its work leaves people and agents able to do next.

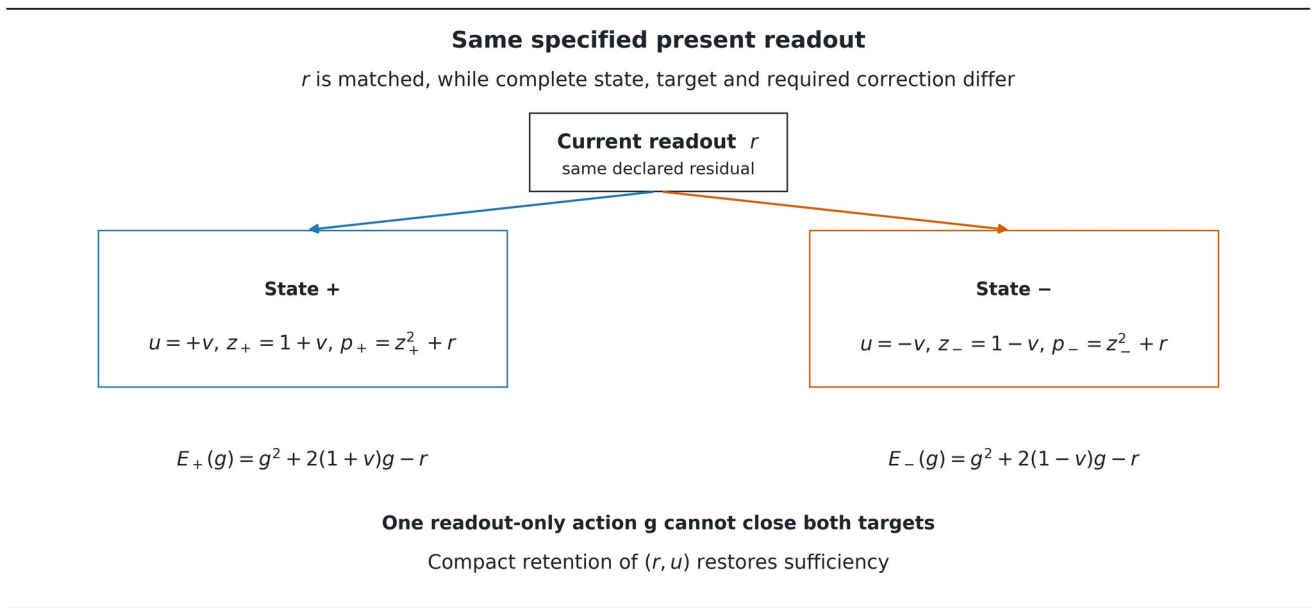
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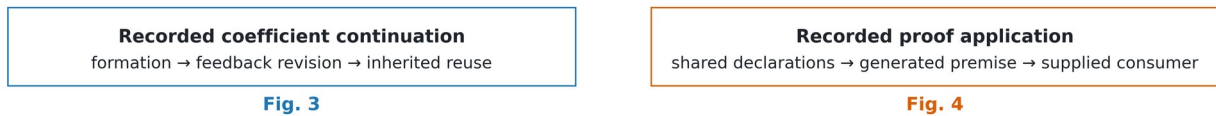
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385 Figure legends

a A specified present readout is not sufficient



b Two recorded entry points into common practice

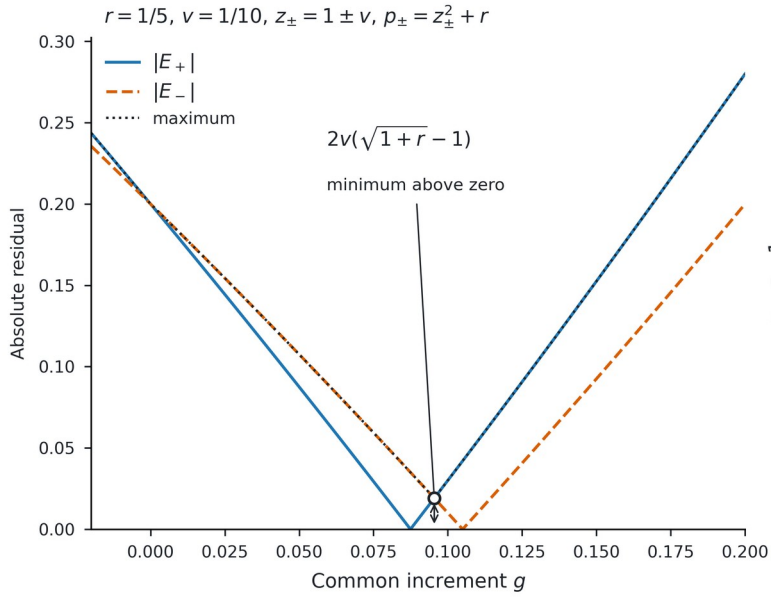
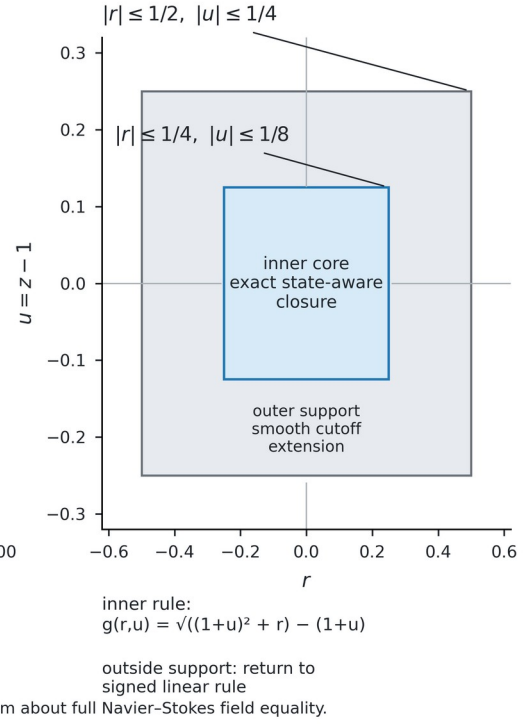


Humans, AI, feedback and adoption remain inside the research object

Evidence identities remain separate: theorem, successor records and Lean proof record are not one identified causal loop

386

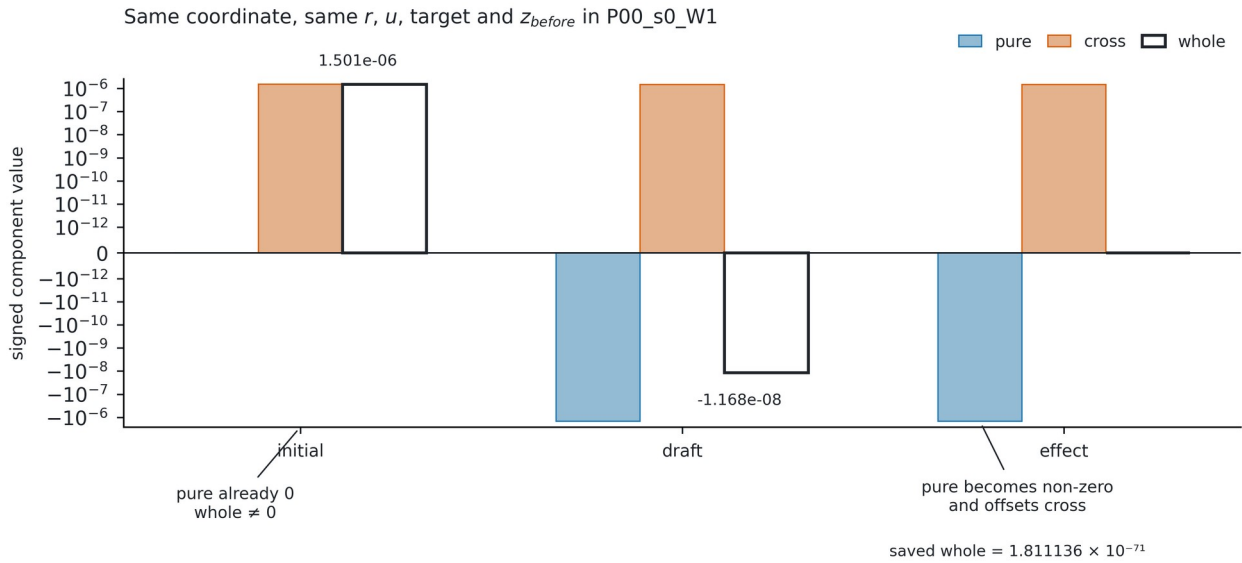
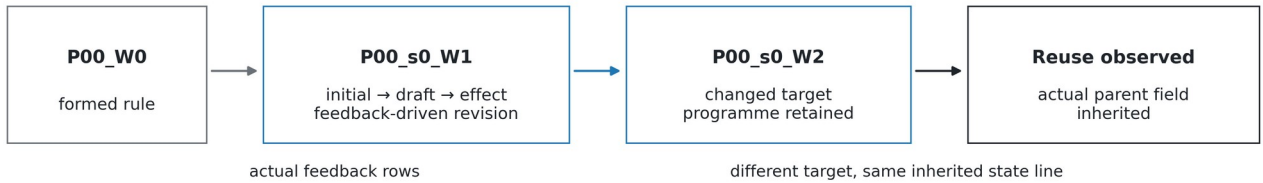
387 **Fig. 1 | A specified current readout is not a sufficient scientific state.** **a**, The analytical witness fixes the declared current
 388 readout r while allowing different complete states and targets. For $0 < r \leq 1/4$ and $0 < v \leq 1/8$, the two states have $u = \pm v$,
 389 $z_{\pm} = 1 \pm v$ and $p_{\pm} = z_{\pm}^2 + r$. A correction that depends only on r must be shared and cannot close both targets. Retaining the
 390 compact pair (r, u) repairs this non-sufficiency. The comparison concerns a specified present readout, not equality of
 391 complete causal states or an identified causal effect of history. **b**, Two archived entry points into common practice are then
 392 separated by evidence identity: coefficient continuation (Fig. 3) and proof application (Fig. 4). The theorem, the successor
 393 records and the Lean proof record are linked conceptually but are not merged into one identified causal loop.

a A positive minimax gap under a readout-only rule**b Compact repair and declared domain**

Analytical curves and declared domains only: no empirical sample size and no claim about full Navier-Stokes field equality.

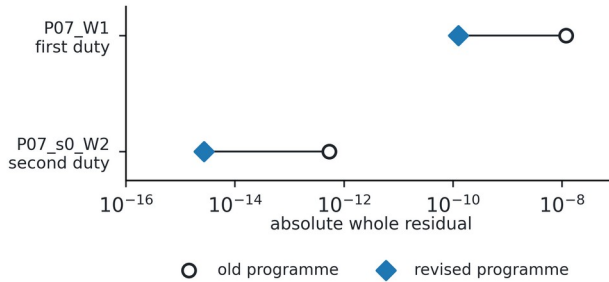
394

395 **Fig. 2 | Positive minimax separation and compact repair.** **a**, Curves show $|E_+|, |E_-|$ and their maximum for the exact
 396 witness at $r = 1/5$ and $v = 1/10$. The minimum of the maximum is the positive distance $2v(\sqrt{1+r}-1)$ above the zero
 397 line. These are analytical values, not experimental samples. **b**, The state-aware rule is exact on the inner rectangle $|r| \leq 1/4,$
 398 $|u| \leq 1/8$. A smooth extension is declared on the outer support $|r| \leq 1/2, |u| \leq 1/4$, and the signed linear rule operates
 399 outside the support. The panel displays domain structure rather than a statistical experiment.

a One revised programme improves the whole relation by giving up a locally exact component**b Actual feedback, parent-state inheritance and changed-target reuse**

$$B_0 = -1/[2(1 + \sqrt{1+r})^2], C_0 = 0$$

final C = smooth secant expression; B retained

c Further revision is compared within, not across, duties

52 recorded stages • 104 complete returns

7/8 adoption pairs kept identical code
• 5 empty final programmes

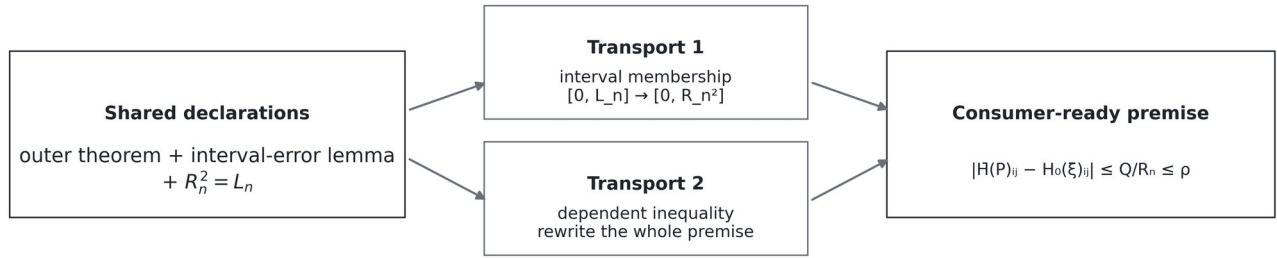
P07 points are post-selected explanatory traces;
they do not establish an average gain.

Do not join different targets into one learning curve.
The two duties use different inherited states and targets.

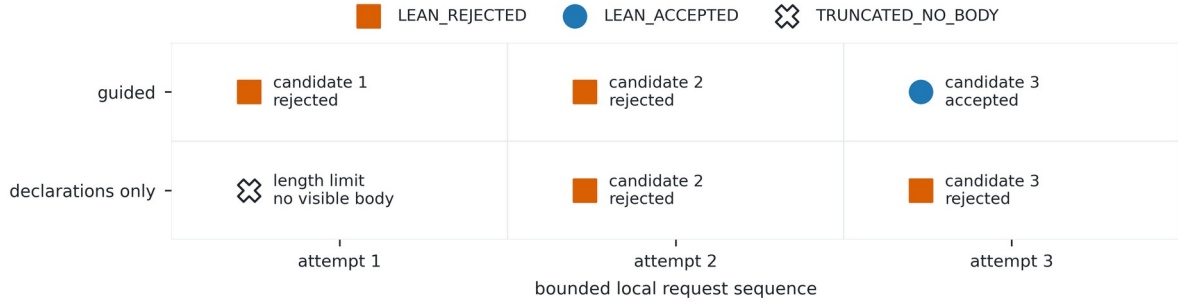
40 parent-state links were checked byte-for-byte.
Unchanged versions, failures and the supplementary
observation remain in Source Data.

400

Fig. 3 | Actual programme revision, inheritance and renewed use. **a**, In assistance episode P00_s0_W1, the same coordinate, r , u , target and z_{before} are shown for the initial, draft and final effect versions. The displayed decomposition is $pure = g^2 + 2g - r$, $cross = 2ug$, and $whole = pure + cross$. Initially, the pure term is exactly zero while the whole residual is not. After revision, the pure term is no longer zero, but it cancels the cross term; the saved final whole residual is retained as 1.811136×10^{-71} rather than replaced by an artificial zero. **b**, The process panel preserves actual feedback-driven revision, programme identity, parent-state inheritance and changed-target reuse from P00_W0 through P00_s0_W1 to P00_s0_W2. **c**, P07 compares old and revised programmes separately within two distinct duties. These are not connected into one learning curve because the duties use different inherited states and targets. The figure retains the full-cohort boundary: 52 recorded stages, 104 completed returns, 7/8 unchanged adoption versions, 5 empty terminal programmes, and the boundary that these post-selected traces do not establish a general average benefit.

a A shared result must be transported into the consumer's representation

Rewriting interval membership alone is not enough; the entire dependent conclusion must also be transported

b The generated proof object is actually used downstream**c All six requests, five candidates and the truncated control request**

First declarations-only request was truncated and had no body; it was not a Lean rejection. Guidance added an application strategy and the downstream consumer was supplied.

This is one selected paired comparison: six requests and five candidates are not independent replications, and saved compilation acceptance is not upgraded here to new kernel certification.

411

412 **Fig. 4 | A generated proof object is made usable and actually used.** **a**, The proof task required two transports from the
 413 common source representation to the consumer's representation: interval membership and the whole dependent inequality, not
 414 interval membership alone. **b**, The generated `hclose` is explicitly distinguished from the supplied downstream consumers
 415 `hmargin` and `ZeroOrderBounds`; the chain `generated hclose` → `supplied hmargin` →
 416 `ZeroOrderBounds` is an observed use relation. **c**, All six requests in the selected paired proof-completion comparison are
 417 shown. The guided condition produced two rejected candidates followed by one recorded acceptance. The declarations-only
 418 condition first terminated at the token limit with no visible proof body; this first control request is therefore truncated rather
 419 than a Lean rejection. The next two declarations-only candidates were rejected. Six requests and five candidates are one
 420 bounded comparison, not independent replications, and saved compilation acceptance is not upgraded here to new kernel
 421 certification.

422 **Methods**

423 **Study basis and evidence scope**

424 This study combines a conditional analytical examination of research continuation with retrospective
 425 analyses of retained coefficient-continuation and proof-completion records. The sources include dated
 426 theoretical and design documents, the original mathematical source, common task materials, generated
 427 candidates, compiler records and prior analyses. The analysis uses the preserved proof-completion
 428 archive, the separate coefficient-continuation archive and independently logged deterministic reanalysis.
 429 It adds no new model responses or experimental units. Newly executed checks are dated and logged
 430 separately from historical compilation and prior audits. Supplementary Notes 1, 5 and 6 distinguish each
 431 evidence identity.

432 **Theory and mathematical scope**

433 A future regime specifies duties, tool semantics, permissions, resource limits and the continuation policies
 434 being compared. Output equivalence is relative to a declared observation M , and future equivalence is
 435 relative to the chosen regime, including the outcome and cost measures. Neither is a claim about complete
 436 causal-state equality. For the finite witness, the observation is precisely r and the restricted policy is a
 437 deterministic real increment depending only on r . We do not assume that a language model obeys that
 438 restriction. The two targets differ because $p_{\pm} = z_{\pm}^2 + r$. The minimax result is an analytic derivation
 439 inherited from the earlier theoretical materials; its full proof and the state-aware sufficient construction
 440 are given in Supplementary Note 2. Symbolic checks and an illustrative deterministic figure are additional
 441 checks; the all-quantifier result is established by the piecewise proof.

442 The smooth cutoff construction has declared inner and outer rectangles. Positivity of the radicand and
 443 denominators establishes analyticity of the specified core functions on a neighbourhood of the outer
 444 rectangle. Transferring this local operation to the source wave classes additionally requires their fixed-
 445 stage, derivative, support and multiplier hypotheses. We do not claim that arbitrary generated expressions
 446 satisfy them, that physical norms improve monotonically, or that this rewrite independently proves a full

NS replacement theorem. Earlier source-relative PDE arguments, recorded computational checks and independently certified mathematics are separate evidence classes. Supplementary Notes 2–3 also state the sufficient-state boundary and its cost assumptions.

Mathematical source and local target

The source is the retained `NavierStokes/PrimaryCovarianceBounds.lean` at the recorded OpenAI repository commit. Its role is a local covariance bound within the published construction, not an original theorem of this study. The fixed outer theorem chooses uniform constants and a scale threshold before the pulse, chart parameter and target multiplier. Its local gap is the assertion that every entry of `normalizedPair P` is within ρ of $H_0 \rho$. The source identifier ρ is written ξ in the paper; `slotRadius r0 h n` is R_n ; `ChartScales.slotLength r0 h n` is L_n ; source coefficients E, D are written e, d ; and c is `PulseCovariance.concentrationConstant a A b B`. The source positivity, compactness, pulse estimates and column-mass results are premises of this task, not model outputs.

Both conditions received the exact outer template with one marked proof-body hole and the full relevant declarations of `normalizedPair_entry_error` and `slotRadius_sq`. The additional note in the guided condition identifies the needed interval/denominator/centre conversion, the integral-error application and the scalar comparison. It does not contain the final tactic body, but it does provide an explicit solution strategy. Full inputs, all five candidate bodies, the truncated-response status and the accepted downstream line are retained in Supplementary Note 5 and the evidence archive. No claim of equal full informational state is made.

Selection, generation and feedback

The final protocol was selected after researcher-directed modification of broader reconstruction workflows. The investigator explicitly authorised a known-proof comparison with the last local step left to the model. Both conditions used a maximum of three requests. The retained seed root was 1,775,273,732, increased by attempt; the recorded arm order was guided then declarations-only. Equal attempt seeds do not imply identical stochastic trajectories under different prompts. Each later request

472 included the condition's own visible candidate and compiler feedback, when available. At most the final
473 12,000 characters of each output stream were exposed as feedback, with truncation marked; complete
474 streams were retained. Models returned Lean proof-body text without live shell, history-retrieval or
475 search tools. The enclosing source was filled mechanically without semantic correction of the generated
476 body.

477 The recorded model identifier was Qwen/Qwen3.8-27B-FP8. The deployment records specify 262,144-token
478 context, one globally active generation, MTP disabled, half-precision computation, tensor parallelism 2,
479 turboquant_k8v4 KV configuration, piecewise capture size 4 and safe synchronisation. Sampling used
480 temperature 1, top-p 0.95, top-k 20, presence penalty 0, repetition penalty 1 and an active end-of-
481 sequence rule, with recorded medium thinking. Per-request output was limited to 32,768 tokens including
482 reasoning; input allowance was 196,608. These are historical configuration records, not independently
483 verified current service properties or model weights. There was no imposed client generation deadline in
484 this final run.

485 **Recorded acceptance and compilation**

486 Five non-empty proof bodies were inserted into the identical template and compiled in the retained Lean
487 4.34.0-rc2 environment. The first declarations-only request had `finish_reason=length` and no visible body,
488 so no compiler was invoked for that response. Recorded compilation took approximately 13–14 seconds
489 per candidate. Non-fatal cache/permission warnings in the saved compiler outputs remain part of the
490 evidence; the logs have not been cleaned into idealised feedback. The successful record contains a zero
491 return code, the exact enclosing type check and the printed axiom dependencies. Failed outputs may list
492 `sorryAx` because their proof remains incomplete; this is not evidence of deliberate use of an axiom to
493 evade the task.

494 Record analysis reread the original handoff and reran the auditor in an isolated output directory. It
495 verified the 859 payload identities, reconstructed all six response streams, matched the five candidate
496 insertions and compiler receipts, and checked the 243 shared work-tree files outside the candidate

497 module. The original collector duplicated two mirrored reasoning fields; their bytes are retained and the
 498 duplication documented. Only visible proof bodies and compiler feedback entered later requests; server
 499 usage is the token source. These checks reproduce recorded acceptance, not a new trusted compilation.
 500 The saved-candidate replay script reports the actual availability of Lean, Lake and the required 4.34.0-rc2
 501 environment; no compatible toolchain was available for this analysis. No new Lean or independent-kernel
 502 validation is claimed.¹⁹

503 Sources and supplementary checks

504 The Codex background refers to the supplied snapshot at commit
 505 a8964cb1bad67bc26a826fb07d1bef99c6a3f008, including the new-context handler and history-notes
 506 extension, backend and tools. The handler requests a new window without summarising the conversation.
 507 The extension exposes note and historical-content operations through a backend; the selected client files
 508 do not establish the server's internal storage or the behaviour of all releases. Public source semantics, our
 509 experimental implementation and opaque internal state are different evidence classes. The mathematical
 510 community's statement supplies a purpose-level argument, not measured general decline, and the NS
 511 process account is attributed to its publisher rather than treated as an isolated causal experiment.

512 Two ancillary records have distinct provenance. The independent diagnostic is a separate Codex task with
 513 its own stated same-start materials and retained local compilation records; it was interrupted and did not
 514 complete the full NS goal. Its model and tool environment differ from the final comparison, and its input
 515 boundary is not independent certification of blindness. The researcher recheck is a reference-exposed
 516 derivation and algebra check; its own account explicitly records no new Lean compilation. Neither record
 517 is pooled with the formal six-request comparison.

518 Evidence interpretation and analysis

519 In the proof-completion study, the unit of observation is one selected obligation and its paired bounded
 520 attempt sequences. The six requests, five candidates, matrix entries and compiler messages are not
 521 independent samples. We report every attempt, the exact endpoint, visible tactic differences and realised

resources. No population success rate, p-value, confidence interval or general efficiency estimate is inferred. The guided prompt adds 332 initial input tokens and procedural information; feedback sequences diverge. Additional explanation, context arrangement, unequal realised effort and stochastic variation remain possible explanations of the observed contrast. Both final candidates locate the relevant lemmas, which supports a specific diagnosis of application failure but does not isolate a unique cognitive mechanism.

The passage from `hclose` to `hmargin` is an observed source-level use of a generated object by a supplied consumer. It does not show that the agent designed the downstream programme, changed its general method or adapted an acquired relation to a new duty. Earlier zero, mixed, stopped and unrun studies retain their original identities in Supplementary Note 6. The separately executed coefficient-continuation evidence is detailed in Supplementary Note 9. Neither its selected favourable traces nor this local proof completion replaces an original study endpoint. Supplementary Note 7 distinguishes observed reconstruction from still-unidentified causal and cross-domain claims.

535 **Archived successor studies**

The assistance study recorded four formation episodes with two shared archived ancestors, followed by two continuations in each of two feedback conditions. The adoption study recorded eight new formation episodes, a common assisted first continuation, and two subsequent version-adoption branches per episode. The totals are 20 and 32 stages. Later targets were fixed modifications of the preceding targets; each continuation read its actual predecessor's amplitude array and programme inventory. The agent wrote two real expressions inside a supplied smooth-cutoff update family. Both branches could retain or reconstruct accessible programmes. These were coefficient-workbench duties, not generation of full NS fields.

The completed-record analysis uses 40 assistance returns and 64 adoption returns. One additional assistance request remains historically unknown; a separately authorised supplementary observation occupies its analysis slot without recovering that request. No unit is added by this editorial reanalysis. The

complete stage counts include one empty final programme in assistance and four in adoption. The original protocol treated these as no applied update, even where a usable draft existed. This historical rule is preserved, not corrected after the fact. Initial and revised code were exactly equal in seven of eight adoption pairs. The original treatment-minus-control family contrasts and complete denominators are reported in Supplementary Note 9 rather than conditioned on valid final outputs.

Two traces, assistance episode P00 and adoption episode P07, are selected after examining the results for their inspectable mathematical content. The programmes, original visible text, feedback rows and parent-state references were reread directly from the archived files. A separate interpreter recomputed their displayed safe-core operations at 100-digit precision, while direct arithmetic checked all 18,720 terminal field rows. Forty inherited-state arrays were matched to predecessor outputs and source hashes. The few displayed coordinates are within-duty illustrations, not independent experimental units or estimates of average improvement. The whole-residual identities are derived analytically from the actual programmes. Their series properties do not license monotone improvement for every source, point or physical norm.

Data availability

Figure Source Data, the complete mathematical task inputs, all five saved proof candidates, six visible proof-completion responses and compilation records, and source-mapped coefficient-continuation records are included in the accompanying review-support archive, MA0FIELD_REVIEW_SUPPORT.zip. The successor material preserves all recorded terminal outcomes and original study denominators, with separately identified explanatory traces. Provenance maps identify every supplied excerpt and any privacy transformation. Unmodified originals are retained in the separate internal evidence archive. Local release candidates additionally preserve the full scientific raw records, with file-level provenance and only necessary non-scientific redactions. This broader sharing scope does not enlarge the analysis inputs, observation units or original denominators. The new manuscript, software and raw records are prepared as three linked Zenodo records; deposition and public release have not occurred. Original data, text and figures are designated CC BY 4.0 and original code Apache-2.0, with third-party rights preserved.

572 **Code availability**

573 The review-support archive contains figure-generation sources, exact mathematical checks, raw-record
 574 reanalysis, the successor-record audit and assembly/replay scripts for the five saved Lean candidates.
 575 REPRODUCIBILITY/README.md identifies the input and output of each offline command and its verification
 576 layer. Default entry points do not contact a model service. Saved-candidate compilation requires Lean
 577 4.34.0-rc2 and the pinned dependencies; it remains unexecuted in this revision, separately from
 578 independent-kernel verification. Third-party notices accompany their sources. The prepared repository
 579 destination is Wangziqu0/MaoField, on a local nature-main candidate that preserves the older main branch
 580 and R3.0 history. A remote branch publication, default-branch switch and GitHub–Zenodo automatic
 581 archive have not been executed; no permanent address for this new code release is asserted.

582 **AI use and research attribution**

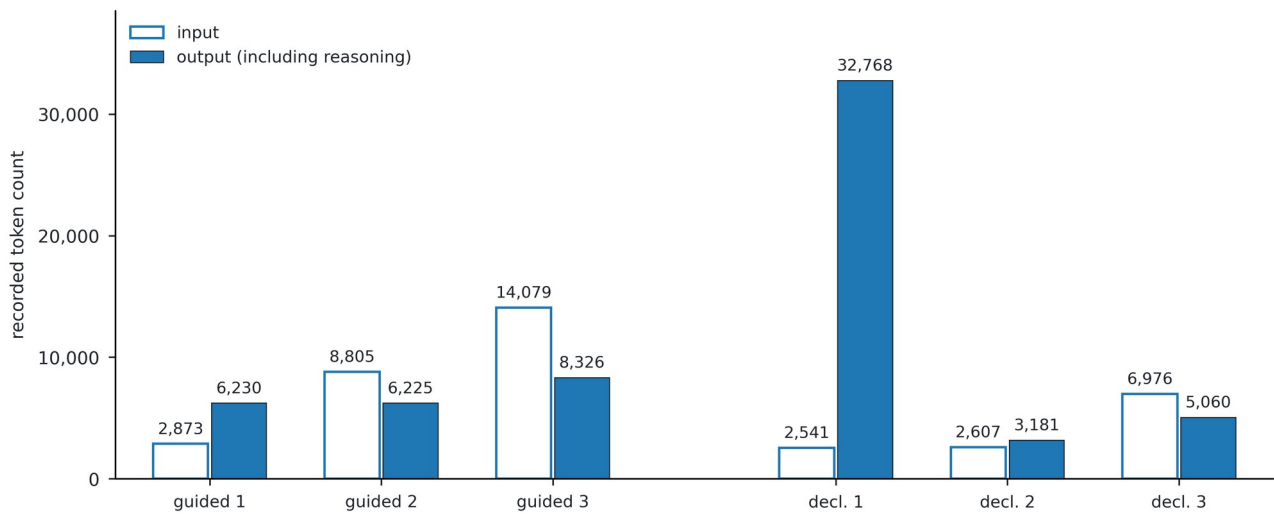
583 The archived studies used model-generated programmes or proof bodies, investigator-designed tasks and
 584 guidance, and Codex-based execution. AI assistance was also used in source organisation, deterministic
 585 analysis, drafting and figure preparation, with these outputs checked against the retained records and
 586 mathematical arguments. This revision adds no new experimental model responses. Source-mathematics
 587 authorship, investigator steering and checking, generated artefacts and computational verification are
 588 distinct contributions. AI systems are not designated as authors.

589 **Additional information**

590 Supplementary Information is available for this paper.

591 **Extended Data**

Extended Data Fig. 1 | Recorded resources in the selected proof-completion comparison



The first declarations-only request was truncated and produced no visible proof body.
Counts describe realised usage only; they are not a general efficiency comparison.

592

593 **Extended Data Fig. 1 | Recorded token use, not efficiency.** The six input and output counts are reproduced from the supplied
594 source-data table. Output includes reasoning. The first declarations-only response reaches its configured limit without a visible
595 proof body. Guided inputs grow with their own candidates and compiler feedback. The figure describes realised work in this
596 one comparison; equal request limits did not equalise realised computation, and no general efficiency advantage is inferred.

597 **Extended Data Table 1 | Earlier workflows retain their identities.** Supplementary Note 6 lists the
 598 retained pre-hclose design history, including stopped or unrun branches. The coefficient-continuation and
 599 proof-completion rows are supported by raw-record reanalysis; older rows retain their historical-report
 600 identity. The completed successor studies are separately identified, not pooled with the proof-completion
 601 pair. Their failed, unchanged, truncated, unknown and NOT_RUN states do not become positive evidence
 602 because the final local completion was accepted.

Recorded workflow	Retained outcome	Evidence status
Full-endpoint workflow	Four branches; 55 requests; no writes	Earlier exploration
Adapted continuation	40 requests; no writes; PI stop	Stopped
Dynamic organisation	23 requests; no writes	Earlier exploration
Material preparation	16 preparation calls; packet not accepted; four branches unrun	Preparation failure
Fixed-material continuation	23 requests; no writes; three remaining branches unrun	Earlier exploration
Precompiled-source continuation	42 requests; no writes; stop	Stopped
Independent Codex diagnostic	Own specified same-start task; saved local compilation; PI stop	Separate diagnostic, not a formal effect control
Researcher recheck	Full reference seen; algebra recheck; no new Lean compilation	Reference-exposed verification
Final hclose pair	Six requests; five candidates; guided third accepted	Present local record
Coefficient assistance / adoption	52 stages; 104 completed returns. Actual revision and reuse; 5 empty finals; 7/8 equal adoption versions.	Executed; raw reanalysis. No established average benefit.